

CUSHING SYNDROME

- Disorder characterized by obesity.
- Hyperactivity of adrenal cortex.

CAUSES

- Due to **hypersecretion of glucocorticoids**, particularly **cortisol**.
- **Classification by origin:**
 - If due to pituitary origin → **Cushing disease**
 - If due to adrenal origin → **Cushing syndrome**

1. Pituitary Origin

↑ sed secretion of ACTH



Hyperplasia of adrenal cortex



Leading to hypersecretion of glucocorticoids

- **ACTH secretion ↑sed by :—**
 - a. Pituitary cells tumor, particularly in basophilic cells which secrete ACTH.
 - b. Malignant tumor of non-endocrine origin like cancer of lungs / abdominal viscera.
 - c. Hypothalamic disorder causing hypersecretion of corticotropin-releasing hormone (CRH).

2. Adrenal Origin

- **Cortisol secretion ↑sed by :—**
 - a. Tumor in zona fasciculata of adrenal cortex.
 - b. Carcinoma of adrenal cortex.
 - c. Prolonged treatment of chronic inflammatory disease (*like rheumatoid arthritis*) with high dose of exogenous glucocorticoids.
 - d. Prolonged treatment with high dose of ACTH, which stimulates adrenal cortex to secrete excess glucocorticoids.

CUSHING SYNDROME CLASSIFIED INTO 2 TYPES

1. ACTH-Dependent Cushing Syndrome

- Due to **hypersecretion of ACTH**.

2. ACTH-Independent Cushing Syndrome

- Secretion of ACTH is **normal**.
- Syndrome develops due to **abnormal membrane receptors** for some peptides like **IL-1**, **Gonadotropin-releasing hormone & Gastric inhibitory polypeptide (GIP)**:
 - ↓ *in cells of zona fasciculata*
 - Peptide bind with abnormal receptors
 - ↓
 - ↑ses secretion of glucocorticoids
 - ↓
 - Result in **Cushing syndrome**.

SIGNS AND SYMPTOMS

- **a. Disproportionate distribution of body fat**, results in abnormal features:
 - **Moon Face**: Edematous facial appearance due to fat accumulation and retention of water & salt.
 - **Torso**: Fat accumulation in chest and abdomen; arms & legs are very slim.
 - **Buffalo Hump**: Due to fat deposit on back of neck & shoulder.
 - **Pot-Belly**: Due to fat accumulation in upper abdomen.
- **b. Purple Striae**: Reddish-purple stripes on abdomen due to:
 - Stretching of abdominal wall by excess subcutaneous fat.
 - Rupture of subdermal tissue due to stretching.
 - Deficiency of collagen fibers due to protein depletion.
- **c. Thinning of extremities**
- **d. Thinning of skin and subcutaneous tissues** due to protein depletion caused by ↑sed catabolism of proteins.
- **e. Acanthosis** :— Darkened skin patches in certain areas (*axilla, neck, groin*).
- **f. Pigmentation of skin** (*in ACTH-dependent Cushing syndrome*).
- **g. Facial Plethora** :— Facial redness.

- **h. Hirsutism** :— Heavy growth of body & facial hair.
- **i. Osteoporosis & bone resorption** —→ Due to protein depletion.
- **j. Hyperglycemia** :— Due to gluconeogenesis —→ Leads to **glycosuria & adrenal diabetes**.
- **k. Poor wound healing**
- **l. Immunosuppression** results in susceptibility for infection.
- **m. Hypertension**
- **n. Weakening of muscles** due to protein depletion.

TREATMENT FOR CUSHING SYNDROME

- Include **cortisol-inhibiting drugs**, **surgical removal** of pituitary or adrenal tumor, **radiation / chemotherapy**.